Family 1.3: family 1.2 plus the stores that could identify plastic in the ocean

a dataset design note — where floating plastic is, and how much

the nets first; the satellites tested against them before anything large is built

blauewelt/earth — design note

11 October 2026

Abstract

Family 1.3 is a proposed *family* — a numbered set of datasets built to one storage design and read by the programme's forecasting model — made of family 1.2 **inherited by reference** (listed in the new family's registry, the JSON file that names every store, with no byte copied) plus a small number of new *stores* (a store is one dataset, published as a small set of arrays) whose job is to say **where floating plastic is in the ocean, and how much**. Plastic is not one of the programme's four prediction goals (El Niño, the ocean carbon sink, the Atlantic overturning circulation and sea-surface temperature), so the family adds a **fifth read-out** — floating plastic concentration at a place and a time — and the stores that measure it, and is kept separate so that family 1.2's registry and its verdicts stay untouched. Every candidate source was measured by request on 2026-10-11 rather than transcribed from its documentation. The finding that shapes everything: **plastic in the open ocean is measured almost only by nets, not by satellites.** A net towed at the surface catches pieces down to a third of a millimetre; a satellite pixel sees a 10 m or 1 km square of sea, and the densest open-ocean microplastic covers about 0.005 % of the surface (Hu 2021). So the first store, `mplastic`, is the nets — built from one producer per cruise, because the published compilations are mostly the same tows compiled again and again; storing pieces per km^2 *as caught*, because the per-volume figures rest on a net depth each compiler assumed; and storing an empty net as the observation it is. The satellite stores are tested against the nets and against human-labelled pixels before anything large is built: Sentinel-2 does not image the open ocean, except in one 2021 campaign over the Great Pacific Garbage Patch that coincides with The Ocean Cleanup's aerial surveys, and a one-hour probe there is expected to find nothing above noise. The note gives the ledger of sources, the stores, nine decisions, the build ladder, the test that can show each store is not worth keeping, and the open items. Nothing has been built, dispatched, rented or uploaded.

1 What family 1.3 is, and the fifth read-out

Family 1.2, the parent. Family 1.2 is the set of global observation stores at 10 km and 5 days or finer — family 1.gf: ocean colour, satellite sea-surface temperature, ship reports, floats, bottles, moorings, the swaths of SWOT (the Surface Water and Ocean Topography satellite, which maps sea level across wide strips) and more — plus the upper-air temperature, humidity and wind of ERA5, the European weather reanalysis (E-085, the experiment that added it). It is published on the project's public Hugging Face dataset `chfrank/earth-tensors`.

Family 1.3. Family 1.2, inherited by reference, plus a small number of new stores whose job is to say where floating plastic is, and how much. Chris asked on 2026-10-08 for this family to be drafted — family 1.2 unchanged, plus stores that could identify plastic in the ocean — and on 2026-10-09 for The Ocean Cleanup's survey to be shown on blauewelt.org first; it shipped that day as the globe's "Floating plastic, North Pacific" layer.

Why a new family and not a new store in 1.2. Plastic is not one of the programme's four prediction goals. Family 1.3 adds a **fifth read-out** (a quantity the model is asked to predict or report): floating plastic concentration at a place and a time, and the stores that measure it. Keeping it a separate family keeps 1.2's registry and its verdicts — the recorded result of each store's keep-or-drop test — untouched.

The finding that shapes everything. Plastic in the open ocean is measured almost only by nets, not by satellites. A net towed at the surface catches pieces down to a third of a millimetre; a satellite pixel sees a 10 m or 1 km square of sea, and the densest open-ocean microplastic covers about 0.005 % of the surface (Hu 2021), far below what any optical sensor can see. Satellites see plastic only where it is piled up — windrows (lines where wind and currents gather floating material), river plumes, harbours — and even then mixed with seaweed and foam. So the first store is the nets, and the satellite stores are tested against them before anything large is built.

2 What the family already holds — and why none of it identifies plastic

Measured from the registries and adapters (the per-source fetch code) on 2026-10-11, these inherited stores bear on plastic:

store (inherited)	what it holds	why it matters for plastic
`oc4k` (1.gf)	daily 4 km ocean colour: log-chlorophyll, light attenuation `kd_490`, observation count, 1997-09 → 2022-12	turbid or bloom water is what a debris index confuses with plastic
`pace4k` (1.gf)	daily 4 km PACE (NASA's 2024 hyperspectral ocean-colour satellite): log-chlorophyll, organic and phytoplankton carbon, apparent visible wavelength, three plankton cell counts, 2024-03 →	the same confusers; **no reflectance and no shortwave-infrared band**
`swot` (1.gf)	2 km swath sea-surface height, radar backscatter `sig0_karin` (linear), uncertainty	currents converge where plastic gathers; backscatter dims over surface films (no published plastic use found)
`swh` (1.gf)	wave height along altimeter tracks	waves mix light plastic below a surface net — the wind-mixing correction
`icoads` (1.gf), `era5_u` / `era5_v` (1.2)	ship winds; upper-air wind on 13 pressure levels	wind drives both the drift and the mixing

Table 1: The inherited stores that bear on plastic, as measured on 2026-10-11.

No inherited channel can identify plastic. Ocean colour at 4 km carries no reflectance bands, and no family-1 store holds a shortwave-infrared band over water. This was found on 2026-10-08 and confirmed against the adapters (`ml/family1/adapters/pace4k.py`, `oc4k.py`, `swot.py`). The inherited stores are therefore context — the confusers, the convergence, the wind — and family 1.3 has to bring its own measurement.

3 The ledger, part one — net tows and water samples, the ground truth

Every row below was measured by request on 2026-10-11 (HTTP status 200 means the file was served in full).

Table 2: Net tows and water samples, as measured on 2026-10-11.

source	access	licence	rows	span	units as published
NOAA NCEI Global Marine Microplastics (the US national archive's harmonised compilation)	ArcGIS FeatureServer (a web map service that answers record queries) `services2.arcgis.com/C8EMgrsFcRFL6LrL/.../Hub_Microplastics_Replace/FeatureServer/0`, and the Hub CSV export of item `1b56a456ad8c4388 9652e62aa25a317e` — 200, identical; the product page names neither	CC0 1.0 (accession 0306475; citation requested, DOI 10.25921/v8gj-k896)	29,076, of which **16,988 ocean-water samples** (893 beach, 580 sediment, 10,615 citizen nurdle counts)	water 1972-04-20 → 2024-10-23	water and beach in pieces/m^3; sediment pieces/kg; nurdles pieces per 10 minutes
The Ocean Cleanup, Lebreton et al. 2018 (figshare 5873142)	figshare API and its download server — 200	CC BY 4.0	683 events + 1,595 aircraft objects + 3,532 compiled tows	2015-07 → 2016-10 (compilation 1972–2015)	pieces/km^2 and g/km^2, four size classes, wind-corrected AND raw
The Ocean Cleanup, Lebreton et al. 2024 (figshare 25674120)	figshare — 200	CC BY 4.0	917 manta tows (510 are the 2015 expedition again, **407 new, 2018–2022**), 162 large-net tows, 74 aerial surveys, 40 cleanup-system catches	2015 → 2022	pieces/km^2 and g/km^2 over 7 size classes
EMODnet Chemistry floating micro-litter v2026 (the EU's marine data network)	`nodc.ogs.it/repository/aggregated-datasets/litter/EMODnet_floating_microlitter_2026_unrestricted.zip` — 200	CC BY 4.0 (in every record)	≈ 3,368 samples (32,067 lines: sample × litter type × size class)	2011-06-14 → 2025-08-30; mostly European seas	**raw counts per net**, with mouth width and either track length (21,497 lines) or filtered volume (10,588) — normalised by the builder
Isobe et al. 2021 (the multilevel dataset behind Japan's Atlas of Ocean Microplastics)	Springer supplement zip — 200	article CC BY 4.0; data licence to confirm	8,218 samples, 27 projects	2000 → 2019	pieces/m^3 (levels 0 and 1); pieces/km^2 wind-corrected (level 2)
AOMI (Japan's Atlas of Ocean Microplastics)	`aomi.env.go.jp/arcgis/rest/services/opendata/MapServer/1000` — 200	**non-commercial only** (the site's data policy); per-record Creative Commons licences, 6,900 rows with none	18,860 raw samples	1972 → 2026-01	pieces/m^3 and pieces/km^2 by size class
Eriksen et al. 2023 compilation ("plastic smog")	raw CSV on GitHub, from the authors' repository — 200	**none in the repository**; the paper is CC BY 4.0	12,546	1971 → 2019	pieces/km^2, g/km^2

Three facts about these sources decide the design.

1. **They are mostly the SAME tows, compiled again and again.** The cruises of SEA (the Sea Education Association), The Ocean Cleanup's 2015 expedition and Eriksen's 5 Gyres tows sit in NCEI, AOMI and Eriksen 2023 alike, and the SEA Pacific tows are in AOMI twice (2,527 submitted by SEA, 2,130 imported via NOAA). 12,160 of AOMI's 18,860 rows match NCEI on date and position; 7,758 of Eriksen 2023's 12,546 do; 644 of the 683 Ocean Cleanup 2015–16 events have an NCEI row on the same day and place. A store that concatenated the compilations would count the same tow three times.

2. **"Pieces/m^3" for a surface net tow is not a measurement.** A neuston or manta net (a surface net that skims the top of the sea) skims the top few tens of centimetres and measures pieces per AREA; NCEI converts to per VOLUME by dividing by a net depth the compiler assumed, and the assumption differs by study. Matched against The Ocean Cleanup's own pieces/km^2, the implied depths are 0.25 m for Law 2014, 0.08 m for Eriksen 2014, 0.155–0.25 m for Goldstein, and 0.075 m for The Ocean Cleanup's 2015 tows; Isobe 2021 uses 0.5 m for the same SEA tows. Values from different studies are not comparable until that division is undone.

3. **Zeros are real and common.** 4,958 of NCEI's 16,988 water samples, 5,566 of AOMI's, 1,506 of

Lebreton's compiled tows: the net came up empty. A zero is an observation and is stored as one.

The SEA conflict — found while checking, settled the same day. SEA's 2,529 Pacific tows (Law et al. 2014) exist in two published copies: NCEI accession 0211008 (also carried by NCEI's harmonised table as 2,530 rows) and the copy inside Lebreton 2018's compilation. They agree on every year-month count and on the first ~ 46 rows, then diverge: only 399 positions match, although both are coherent ship tracks (median step between consecutive tows 92 km and 100 km). Two further copies broke the tie. SEA's own submission to AOMI (2,485 Pacific tows, 2001-10 $\rightarrow$ 2012-12) matches Lebreton's copy at **all 2,485** year-month-position keys and NCEI's at 355; Eriksen 2023 (10,170 SEA rows carrying ship, cruise and station numbers) matches Lebreton's at 2,485 of 2,487 and NCEI's at 357. **NCEI accession 0211008 has the wrong positions; Lebreton's copy — the one the globe's "earlier surveys" option draws — is right.** In September 2005, for example, NCEI places the tows off Hawaii (15° N, 160° W) where the other three place them off San Diego (32.8° N, 117.5° W). Only the Pacific accession is affected: NCEI's 6,135 SEA Atlantic tows (Law et al. 2010) match SEA's own AOMI submission at 5,808 of 5,851 positions.

The values are a separate, open question. Where the positions agree, the published Law 2014 value equals SEA's own AOMI value (pieces/km^2, < 5 mm) for 1,440 of 2,487 tows and is within 5 % for 20 more; for the other 1,027 SEA's resubmitted value is larger, typically about three times (median ratio 3.0, 10th–90th percentile 1.02–5.85). That looks like a later reprocessing by SEA, not a transcription error, and nothing in the files says which version SEA now stands by. Decision D9 records the choice.

4 The ledger, part two — satellites, and the one window where they meet the nets

Table 3: Satellite sources and labelled sets, as measured on 2026-10-11. 200 and 206 are the HTTP codes for a whole file and for a requested byte range served; 302 $\rightarrow$ 401 is a redirect to a login that refused.

source	what it is	access	licence	the measured fact that matters
Sentinel-2 MSI (the MultiSpectral Instrument on the EU's Sentinel-2 satellites)	10–20 m optical imager, 13 bands including red-edge B06 (740 nm), near-infrared B08 (842 nm), and shortwave-infrared B11 (1,610 nm) and B12 (2,190 nm)	level-2A COGs on AWS (`sentinel-cogs`): anonymous range read **206**, CORS ∗; level-1C as JPEG 2000 in `sentinel-s2-l1c`, anonymous 206 from here (documented as requester-pays — recheck outside the proxy)	Copernicus free, full and open; notice "Contains modified Copernicus Sentinel data [year]"	**It does not image the open ocean.** ESA's acquisition plan covers land, coastal waters to 20 km, islands, the Mediterranean and closed seas; the measured offshore reach is ~ 250 km where a swath clips land (off Portugal: items at 65–250 km, none at 350 km). The Great Pacific Garbage Patch box (30–35° N, 135–145° W) has **0 items in 2023 and in every year except one special campaign: 2,602 level-2A items, 61 dates, 2021-07-02 $\rightarrow$ 2021-09-19**
… its level-2A over water	Sen2Cor's atmospheric correction is tuned for land	measured on one clear tile — 100+ km off Portugal	—	**11 % of B08, 27 % of B11 and 8 % of B06 pixels pinned at the lowest value** — the bands the debris index differences. The debris papers start from level-1C and correct with ACOLITE (an open-source atmospheric correction tuned for water)
MARIDA (Marine Debris Archive, Kikaki et al. 2022)	human-labelled Sentinel-2 pixels in 15 classes	Zenodo 5151941 — 200, 1.16 GB	CC BY 4.0	1,381 patches of 256 × 256 pixels, 63 scenes, 2015-11 $\rightarrow$ 2021-01, 11 countries; **837,377 labelled pixels, 3,399 of them "marine debris"** (in 373 patches)
MADOS (Kikaki et al. 2024)	the same kind; adds oil spills	Zenodo 10664073 — 200, 4.04 GB	CC BY 4.0	174 scenes, 2,803 crops of 240 × 240, 2015–2022
Plastic Litter Project (Topouzelis, University of the Aegean)	KNOWN plastic targets moored in the Gulf of Gera, Lesvos, imaged by Sentinel-2 and drones	Zenodo 3752719 (2019), 7085112 (2021, 159.5 GB), 10046182 (2022–23) — 200	CC BY 4.0	2021: 22 acquisition dates, 2021-06-11 $\rightarrow$ 2021-10-04; detection of targets covering ≥ 25 % of a pixel shown (Topouzelis 2020)
FloatingObjects (Mifdal et al. 2021)	line annotations of floating objects on 26 scenes	Google Drive zip, 16.6 GB — 206	Apache-2.0	2,270 features, 2018–2020

source	what it is	access	licence	the measured fact that matters
PACE OCI level 2	the 2024 hyperspectral ocean-colour instrument (the Ocean Color Instrument on NASA's PACE satellite) at ∼ 1 km	CMR (NASA's catalogue) anonymous; downloads need an Earthdata login (302 → 401); the anonymous OB.DAAC OPeNDAP server holds only levels 3 and 4	NASA open	water-leaving reflectance stops at 719 nm; the only shortwave-infrared above level 1 is the **surface-reflectance product SFREFL, `rhos` at 1038, 1249, 1618, 2131 and 2258 nm**; **no band sits on plastic's 1,732 nm absorption**; ∼ 89 GB a day at level 2; anonymous only as level-3 2 km / 4 km daily maps
CYGNSS microplastic level 3 (Evans & Ruf 2021)	an empirical index from the sea-surface roughness that CYGNSS measures in reflected GPS signals, fitted to a plastic MODEL	PO.DAAC, Earthdata login	NASA open	0.25° daily files, but the authors' feature resolution is 1° and 30 days; latitudes within about ±38°; confounded by natural surfactants and the tropical rain band

Sentinel-2's coverage. The one satellite with the bands the debris indices need (red edge, near infrared, shortwave infrared) at 10–20 m does not image the open ocean: its plan stops at coastal waters, islands and closed seas, and the measured reach offshore is about 250 km and only where a swath clips land. Over the Great Pacific Garbage Patch it imaged one campaign in one summer, 2021.

Level-2A clips over water. The anonymously served, already-corrected product cannot be used as it is: on a clear tile more than 100 km off Portugal, 11 % of B08, 27 % of B11 and 8 % of B06 pixels sit at the lowest value — the very bands the floating debris index differences. Any debris store starts from level-1C and corrects with ACOLITE, as the debris papers do.

The labelled sets. MARIDA, MADOS, the Plastic Litter Project and FloatingObjects are human labels or known targets on Sentinel-2 scenes, all near coasts. They are what any satellite debris method is tested on, and the first three are CC BY 4.0.

PACE and CYGNSS. PACE's water-leaving reflectance stops at 719 nm, its only shortwave-infrared above level 1 is the surface-reflectance product, and no band sits on plastic's 1,732 nm absorption. CYGNSS's microplastic index is a roughness anomaly fitted to a plastic model, at 1° and 30 days, and natural surfactants and the tropical rain band confound it.

The 2021 co-location window — where a satellite and the nets meet in the open gyre. The Ocean Cleanup's North Pacific missions of 2021 (in Lebreton 2024) flew 22 aerial surveys (34.8–35.4° N, 144.4–145.9° W, every few days from 2021-08-05 to 08-31) and towed 6 manta nets in August, inside the area and the weeks of the only Sentinel-2 campaign over the patch: a 0.7° box around the surveys has 167 level-2A items on 20 dates in August–September 2021, including 2021-08-11, 08-16, 08-24, 08-26 and 08-31 — days with surveys. It is the one place on Earth where an open-gyre plastic measurement and a 20 m multispectral image are days apart, and it is cheap to look at (§7, step 2).

5 The stores — what each holds, the indices it would compute, and the footprints

Table 4: The proposed stores. The first three are built in phase A.

store	tier	what one row / frame is	channels	rows / size	phase
`mplastic`	P (points)	one surface sample: a net tow, a pump or grab sample, or an aircraft survey line	`n_km2` (pieces/km^2, as caught), `g_km2`, `n_m3` (volumetric samples only), `n_km2_wc` (the producer's wind-mixing-corrected value, where given), four size-class counts where the producer resolves them (0.05–0.5, 0.5–5, 5–50, > 50 cm), `mesh_um` (net mesh in μm), `depth_m` (net depth, assumed or measured), `wind_ms` at sampling; `qc` = gear class (manta, neuston, large net, pump, grab, aircraft); `platform` = cruise	≈ 20,000 rows after de-duplication (NCEI ≈ 17 k water samples, of which ≈ 0.6 k are replaced by The Ocean Cleanup's own ≈ 1.2 k rows, + EMODnet ≈ 3.4 k − ≈ 0.6 k already in NCEI); < 5 MB	A

store	tier	what one row / frame is	channels	rows / size	phase
`mobjects`	P	one object identified in an aircraft photo mosaic	length, width, top-view area, weight low / mid / high; `qc` = object type (net, rope, buoy, container, unknown . . .)	1,595 (2016) + the 2021–22 surveys if Lebreton 2024 resolves objects; < 1 MB	A
`s2labels`	P	one human-labelled Sentinel-2 pixel	the 11 ACOLITE-corrected band reflectances the label set ships; `qc` = class (marine debris, *Sargassum*, foam, ship, clear water, turbid water, . . .), confidence	MARIDA 837,377 (measured) + MADOS ≈ 1.5 M (its project page) + the Plastic Litter Project's target pixels (to count) — ≈ 2.4 M rows, ≈ 130 MB at 53 bytes a row	A
`s2fdi_gpgp`	G-fine	one 20 m Sentinel-2 frame over the patch, July–September 2021	FDI, FAI, NDVI, PI and the four bands they use, from level-1C via ACOLITE	only if step 2 passes; the campaign is 2,602 tiles × $5,490^2$ pixels × 8 float16 channels ≈ 1.25 TB before any cut to water and cloud-free	B
`s2fdi_coast`	G-fine	the same over the water of the coastal band and enclosed seas	as above	a cost estimate, not a build	C
`pace_swir`	G	PACE `rhos` at 865, 1038, 1249, 1618, 2131, 2258 nm, daily 4 km (anonymous level 3) or 1.2 km (level 2, Earthdata)	6 reflectances	≈ 2 TB a year at level 2	C
`cygnss_mss`	P or G	GPS-reflection roughness, the observable under the CYGNSS index	mean-square slope, wind	to measure	C
`beachlitter`	P	shoreline litter surveys (NCEI beach and nurdle counts, EMODnet beach litter, NOAA MDMAP)	counts per transect or per 10 minutes	≈ 12 k NCEI rows + others	C

The indices, as their papers define them. R_n is the reflectance in Sentinel-2 band n and λ_n that band's centre wavelength (band 4 is red, 6 red edge, 8 near infrared, 11 shortwave infrared).

- **FDI**, the floating debris index (Biermann et al. 2020), brightens where something floats on the water: it is band 8 minus a baseline drawn from bands 6 and 11,

$$\mathrm{FDI} = R_8 - R'_8, \qquad R'_8 = R_6 + (R_{11} - R_6)\,\frac{\lambda_8 - \lambda_4}{\lambda_{11} - \lambda_4} \times 10,$$

  with the centre wavelengths of each satellite: the factor $10\,(\lambda_8 - \lambda_4)/(\lambda_{11} - \lambda_4)$ is 1.7722 for Sentinel-2A and 1.7770 for Sentinel-2B.

- **FAI**, the floating algae index (Hu 2009):

$$\mathrm{FAI} = R_8 - \left[R_4 + (R_{11} - R_4)\,\frac{\lambda_8 - \lambda_4}{\lambda_{11} - \lambda_4}\right].$$

- **NDVI**, the normalised difference vegetation index, and **PI**, the plastic index (Themistocleous et al. 2020):

$$\mathrm{NDVI} = \frac{R_8 - R_4}{R_8 + R_4}, \qquad \mathrm{PI} = \frac{R_8}{R_8 + R_4}.$$

Footprints. Every value reaches the model with a footprint, the area and time span it stands for. A manta tow sweeps $\sim 0.9\,\mathrm{m} \times 1$–$15\,\mathrm{km}$: its token's area is the swept area (≈ 0.001–$0.02\,\mathrm{km}^2$) and its length the tow line; a pump sample is a point (`log2_fp` $= -4$, the family's label for a point); a Sentinel-2 pixel is 20 m.

6 Decisions D1–D9 — this note's picks, each reversible

D1 The first store is the nets. `mplastic` is built first; every satellite store is gated on a test against it or against labelled pixels.

D2 One producer per cruise, the original producer first. Precedence: The Ocean Cleanup (its own cruises, its own size classes) → NCEI (each study's own accession where one exists, else the harmonised table) → EMODnet (European cruises) → Isobe 2021 (rows not in the above, once its data licence is confirmed). A row from a later source is dropped when an earlier one holds the same day within 0.01°, and its drop is counted in the de-duplication ledger, never silently. Compilations are sources of rows nobody else published, not of copies.

D3 Store what the net measured. For surface net tows the stored quantity is pieces/km^2 (and g/km^2 where weighed) **as caught**; NCEI's pieces/m^3 is multiplied back by the depth the compiler assumed, recorded per study in a committed table with the match that measured it. Pumps and grab samples are volumetric and stay in pieces/m^3, in their own channel. The producer's wind-mixing correction is kept beside the raw value, never in place of it.

D4 AOMI stays out of phase A: non-commercial, mixed per-record licences, 64 % already in NCEI, and 79 rows with latitude and longitude swapped. The probe measures what it would add; if it is worth it, it goes on the *private* track (held for this project's own use, never republished).

D5 Eriksen 2023 stays out until its licence is clear (there is no licence in the repository). It was one of the two further copies that settled the SEA conflict, a use that needs no redistribution.

D6 No satellite debris store is built at scale in this family's first pass. Sentinel-2 cannot see the open ocean, PACE has no band on plastic's absorption, and CYGNSS's index is fitted to a model. What IS built is cheap: the labelled pixels (`s2labels`) and one probe at the 2021 co-location window. A coastal Sentinel-2 debris store is a phase-C decision with a cost estimate, taken on the probe's result.

D7 CYGNSS's microplastic index is rejected under the family's rule — raw observations first, a derived product only where it carries observations nothing else does: it is a roughness anomaly fitted to a plastic model, at 1° and 30 days. The roughness itself (mean-square slope) is a phase-C candidate.

D8 Family code `13`, version `1.3`, root `tensors/family1_3/`, registry `tensors/family1_3/family13.json`. It inherits 1.2, which inherits 1.gf: the registry's `inherits_block` lists the RESOLVED chain (1.gf's 13 stores under their own paths and 1.2's four), so a reader opens one parent list, not a chain. `ml/build_family1_registry.py` needs that small extension; today it resolves one parent.

D9 SEA's Pacific tows come from Lebreton 2018's copy, with the published Law et al. 2014 values. NCEI's accession 0211008 and its 2,530 "Law et al. 2014" rows are dropped by name (their positions are wrong); SEA's Atlantic tows (Law et al. 2010) stay with NCEI. The ledger records, per SEA Pacific tow, whether SEA's later AOMI value differs, so the choice can be reversed without rebuilding anything else. Whether SEA's resubmitted values supersede the published ones is a question for SEA.

7 The ladder — the order of building

0 Settle the SEA conflict (§3) — **done 2026-10-11**: SEA's own AOMI submission and Eriksen 2023 both agree with Lebreton's copy and not with NCEI's accession 0211008. The globe layer keeps Lebreton's copy; `mplastic` drops NCEI's SEA Pacific rows (D9). Left open: which values SEA now stands by.

1 Phase A, on a hosted runner (one of GitHub's own build machines), $0. Adapters for `mplastic`, `mobjects` and `s2labels` against the family-10 point-store builder; a probe that runs the whole of each (they are small enough that the probe IS the build); the per-study depth table committed with the match that measured each depth; the de-duplication ledger (rows read, rows dropped as copies per pair of sources, rows kept). Publish to `tensors/family1_3/`, registry `family13.json` inheriting the resolved 1.2 chain.

2 The co-location probe — one hour, $0. For each 2021 aerial survey and manta tow in the patch with a Sentinel-2 acquisition within ±1 day, fetch the level-1C tiles over the survey line, correct with ACOLITE, compute FDI, and compare FDI's distribution under the line (and its upper percentiles) with the same tile's open water away from it, and with the survey's megaplastic count. **Expected: nothing above noise** — the densest patch holds at most ∼ 13 objects over 50 cm per km^2 (Lebreton

2018's highest mosaic), of the order of 0.001 % of the surface. A clear signal would be a finding worth a paragraph; no signal closes `s2fdi_gpgp` for good, with a measured reason.

3 **Phase B** only on step 2's result: `s2fdi_gpgp`, if it showed signal.

4 **Phase C** decisions, each with its own probe first: `s2fdi_coast`, `pace_swir`, `cygnss_mss`, `beachlitter`, and AOMI's unique remainder on the private track.

8 The falsifiers — what would show each store is not worth keeping

`mplastic` **as a read-out.** The family is worth having only if a model reading 1.3's inputs predicts held-out tows — left out by CRUISE, so a cruise's neighbouring tows cannot leak — better than two baselines: the nearest tows in space and time (k nearest neighbours, $k = 5$), and the published climatological map (Isobe 2021's level-3 grid). Score: the error in $\log_{10}(1 + \text{pieces/km}^2)$ and, separately, whether an empty net is predicted empty. If the model does not beat the neighbours, the inputs carry nothing the tows' own geography does not, and the read-out is shelved.

`mobjects`**.** Kept as the aircraft's own record; falsified only if its counts do not reproduce Lebreton 2018's published megaplastic concentrations per mosaic — a check made when the store is baked.

`s2labels`**.** The baseline is an index threshold — FDI above its 95th percentile over clear water in MARIDA's training split — scored on MARIDA's own test split. The store earns its place if a model reading the 11 bands plus the inherited ocean colour beats it on debris-versus-not F1 (a score that balances missed debris against false alarms). If the threshold already matches the model, the labels are a test set, not a training store.

`s2fdi_gpgp`**.** The co-location probe of §7, step 2.

9 Open items, in the order they block

Status: **draft — nothing built.** The sources were measured on 2026-10-11; the raw downloads are in scratch space, not committed.

1. settled 2026-10-11: the SEA Pacific positions (NCEI's accession is wrong, D9). Still open, and not a blocker: whether SEA's resubmitted values (about 3× for 1,027 tows) supersede the published ones — a question for SEA;
2. Isobe 2021's data licence; Eriksen 2023's licence;
3. whether the repository's Earthdata account (secrets `EARTHDATA_USERNAME` / `EARTHDATA_PASSWORD`, working for LP DAAC, GES DISC and PO.DAAC) reads OB.DAAC's PACE level 2 — never tried;
4. the registry builder's resolved-chain inheritance (D8);
5. the level-1C bucket's requester-pays status outside this machine's network proxy.

Terms used above, in one line each

Family a numbered set of datasets built to one storage design and read by the programme's forecasting model; **store** one dataset of a family; **registry** the JSON file listing every store of a family. **Family 1.gf / 1.2 / 1.3** the global observation stores at 10 km and 5 days or finer; 1.gf plus ERA5's upper air; 1.2 plus the plastic stores. **Inherited by reference / `inherits_block` / resolved chain** a parent's stores listed in the child's registry without copying a byte; the registry field holding that list; the list written out in full so a reader opens one parent list, not a chain. **Read-out** a quantity the model is asked to predict or report. **Verdict** a store's recorded keep-or-drop result. **Tier P / G / G-fine** points, profiles and tracks; gridded fields; grids finer than 0.25° stored as compressed tiles. **Footprint / `log2_fp` / token** the area and time span a value stands for; its area as a base-2 logarithm, −4 being the label for a point; the unit the model reads. `qc` / `platform` a point store's quality-or-class column and its instrument (here cruise) column. **Phase A / B / C** the cheap first build,

the stores taken on a probe's result, and decisions taken later. **Probe** a test run of a store's real fetch code before any full build. **Falsifier** the test, fixed before building, that can show a store is not worth keeping. **Hosted runner** one of GitHub's own build machines. **Public / private track** a store anyone may read, or one held for this project's own use because its licence forbids republishing. **Adapter** the per-source fetch-and-convert code. **Hugging Face / `chfrank/earth-tensors`** the dataset host and the project's public repository there. **E-085** the experiment that added ERA5 to family 1.2. **ERA5** the European weather reanalysis, a weather model run over past decades and continually corrected towards observations. **SWOT / `sig0_karin`** the Surface Water and Ocean Topography satellite and its radar backscatter. **PACE / OCI** NASA's Plankton, Aerosol, Cloud, ocean Ecosystem satellite (2024) and its Ocean Color Instrument, which resolves the whole visible spectrum (hyperspectral). **Ocean colour / `kd_490`** the colour of light leaving the sea, from which chlorophyll is estimated; the attenuation of light at 490 nm. **Altimeter** a radar satellite measuring sea height and wave height straight below it. **Wind-mixing correction** the producer's scaling up of a net's catch for light pieces that waves pushed below the net. **Confusers** turbid water, blooms, *Sargassum* (a floating brown seaweed) and foam, which a debris index can mistake for plastic. **Windrows** lines where wind and currents gather floating material. **Manta / neuston net** surface nets that skim the top few tens of centimetres of the sea; **large net** a bigger net towed for larger pieces; **pump / grab sample** water pumped or scooped and filtered, a per-volume measurement. **Mesh** the size of a net's openings (μm, micrometres). **Microplastic / megaplastic** the smallest size class (here 0.05–0.5 cm) and objects over 50 cm. **Nurdles** plastic pellets, the raw material of plastic goods, counted on beaches. **Compilation** a table gathered from many studies rather than measured by its publisher. **Accession / DOI** an archive's catalogue number; a permanent identifier for a dataset. **CC0 / CC BY 4.0 / non-commercial / Apache-2.0** public domain; free with attribution; free only for non-commercial use; a permissive software licence. **NOAA / NCEI** the US National Oceanic and Atmospheric Administration and its National Centers for Environmental Information, the US national environmental archive. **ArcGIS FeatureServer / MapServer / Hub** web services that answer queries for map records. **CSV** comma-separated values, a plain-text table. **figshare / Zenodo / Springer supplement / GitHub / Google Drive** the public file hosts the sources come from. **The Ocean Cleanup** the organisation whose surveys produced Lebreton 2018 and 2024; **cleanup-system catches** what its collecting systems brought up. **EMODnet** the European Marine Observation and Data Network. **AOMI** Japan's Atlas of Ocean Microplastics; **Isobe levels 0–3** its source dataset's stages: 0 and 1 per volume, 2 per area and wind-corrected, 3 a gridded map. **SEA / 5 Gyres** the Sea Education Association, whose cruises towed the Pacific nets of Law 2014, and 5 Gyres, the organisation behind Eriksen's tows. **NOAA MDMAP** NOAA's Marine Debris Monitoring and Assessment Project of beach surveys. **Great Pacific Garbage Patch** the area of the North Pacific gyre where floating plastic gathers. **Sentinel-2 / MSI / S2A, S2B** the EU Copernicus optical satellites and their MultiSpectral Instrument; the two satellites. **Red edge / NIR / SWIR** light just beyond red, near-infrared, shortwave-infrared. **nm** nanometre, the unit of wavelength. **Level 1C / 2A** Sentinel-2's top-of-atmosphere product and its surface product after atmospheric correction; **level 2 / 3 / 4** elsewhere: swath measurements, gridded maps, model-filled products. **Atmospheric correction / Sen2Cor / ACOLITE** removing the air's contribution to what a satellite sees; ESA's land-tuned corrector; an open-source corrector tuned for water. **Item / tile / scene** one catalogued Sentinel-2 image file, its fixed grid square, one acquisition. **COG / JPEG 2000** image formats; a cloud-optimised GeoTIFF can be read in byte ranges. **AWS / requester-pays** Amazon Web Services' storage; a bucket whose downloads are billed to the reader. **HTTP 200 / 206 / 302 / 401** whole file served; byte range served; redirect; login refused. **CORS `*`** a header allowing any website to read the file. **Copernicus / ESA** the EU's Earth-observation programme and the European Space Agency. **MARIDA / MADOS** the Marine Debris Archive and its successor with oil spills, labelled Sentinel-2 pixels. **Plastic Litter Project** the University of the Aegean's moored plastic targets off Lesvos. **FloatingObjects** line annotations of floating objects on Sentinel-2 scenes. **CMR / Earthdata login / OB.DAAC / PO.DAAC / LP DAAC / GES DISC** NASA's catalogue, its free user account, and its ocean-biology, ocean, land and Earth-science data centres. **OPeNDAP** a protocol for reading subsets of scientific files over HTTP. **SFREFL / `rhos`** PACE's surface-reflectance product and its reflectance variable. **1,732 nm absorption** the wavelength at which plastic absorbs light distinctively. **CYGNSS / mean-square slope** NASA's Cyclone Global Navigation Satellite System, small satellites that measure sea roughness from reflected GPS signals; the roughness they measure. **Surfactants** natural films that smooth the sea surface. **FDI / FAI / NDVI / PI** the floating debris, floating algae, normalised difference vegetation and plastic indices of §5. **Float16** a two-byte number. **Percentile** the value below which a given share of the data lies. k **nearest neighbours** a forecast that averages the k closest observations. **Training / test split** the labelled data a method is tuned on, and the part kept back to score it. **F1** a score combining missed detections and false alarms. **TB / GB / MB** tera-, giga- and megabytes.

References

[1] Biermann et al. (2020). The floating debris index (FDI).

[2] Eriksen et al. (2014). Net tows, implied net depth 0.08 m in NCEI.

[3] Eriksen et al. (2023). The “plastic smog” compilation, 12,546 rows; no licence in the authors’ repository, paper CC BY 4.0.

[4] Evans & Ruf (2021). The CYGNSS microplastic level-3 index. PO.DAAC.

[5] Goldstein. Net tows, implied net depth 0.155–0.25 m in NCEI.

[6] Hu (2009). The floating algae index (FAI).

[7] Hu (2021). The densest open-ocean microplastic covers about 0.005 % of the surface.

[8] Isobe et al. (2021). The multilevel dataset behind Japan’s Atlas of Ocean Microplastics. Springer supplement; article CC BY 4.0.

[9] Kikaki et al. (2022). MARIDA, the Marine Debris Archive. Zenodo 5151941, CC BY 4.0.

[10] Kikaki et al. (2024). MADOS. Zenodo 10664073, CC BY 4.0.

[11] Law et al. (2014). SEA’s Pacific tows; NCEI accession 0211008.

[12] Lebreton et al. (2018). The Ocean Cleanup’s 2015–16 survey. figshare 5873142, CC BY 4.0.

[13] Lebreton et al. (2024). The Ocean Cleanup’s surveys 2015–2022. figshare 25674120, CC BY 4.0.

[14] Mifdal et al. (2021). FloatingObjects. Apache-2.0.

[15] NOAA NCEI. Global Marine Microplastics. Accession 0306475, DOI 10.25921/v8gj-k896, CC0 1.0.

[16] Themistocleous et al. (2020). The plastic index (PI).

[17] Topouzelis et al. (2020). The Plastic Litter Project. Zenodo 3752719, 7085112, 10046182; CC BY 4.0.